

Quantum Engineering Curriculum

Sprint 3 Quiz

Instructions: Answer without running code unless the question explicitly asks you to. Explain your reasoning where appropriate.

1. What does the Pauli-X gate do?
2. Why is X considered the quantum analogue of a NOT gate?
3. Predict the result of X followed by X, then measurement.
4. Why does measuring twice into the same classical bit overwrite the earlier result?
5. Why does `qc.x(1)` raise `CircuitError` when using `QuantumCircuit(1,1)`?
6. Briefly compare deterministic and probabilistic circuits.